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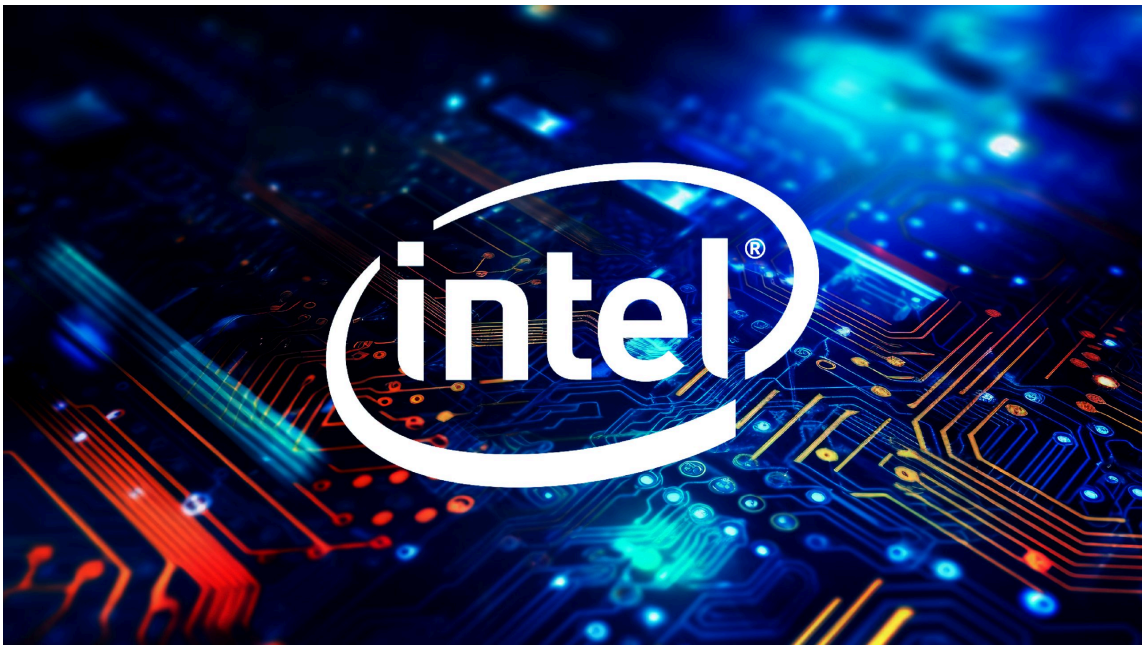

iGPT

Architecture Overview

Intel Generative Pre-trained Transformer Platform

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1. Executive Summary

Purpose

This document defines the **iGPT AI Architecture** for Intel's internal enterprise Generative AI platform. iGPT provides employees with conversational AI capabilities, custom assistant creation, and a marketplace ecosystem for sharing AI solutions across the organisation.

Scope

The architecture encompasses three core capabilities:

- **Chat Interface** – Central conversational AI for multi-turn interactions with enterprise LLMs (GPT-4, GPT-3.5, and other models)
- **Assistant Builder** – No-code wizard for creating, configuring, and publishing custom AI assistants with model selection, system prompts, temperature control, and live testing
- **Marketplace** – Enterprise-wide assistant store for discovering, sharing, and reusing AI assistants across Intel

Key Architectural Components:

- **iGPT Web Application** – Primary entry point at igpt.intel.com/chat; responsive SPA with sidebar navigation, chat workspace, and builder panels
- **LLM Gateway** – Model routing layer supporting GPT-4, GPT-3.5, and other enterprise LLMs with temperature and parameter controls
- **RAG Engine** – Retrieval Augmented Generation for document-grounded responses using uploaded files and data pipelines
- **Intel SSO / Azure AD** – Enterprise authentication via Single Sign-On with Azure Active Directory integration
- **AGS Entitlements** – Fine-grained access control for Secure Group sharing of assistants
- **Marketplace Service** – Discovery, publishing, and catalog management for shared assistants

> **InfoSecurity Restriction:** Assistants with attached files can only be published as Private. This is a directive from InfoSecurity, not a technical limitation.

Key Architectural Principles

Principle	Description
Self-Service AI	Any Intel employee can use conversational AI and build custom assistants without code
Private by Default	All new assistants default to owner-only access; compliance gate for broader sharing
Multi-Model Flexibility	Users select from multiple LLM backends (GPT-4, GPT-3.5, others) per use case
Knowledge Augmentation	RAG-powered document grounding enables context-aware, enterprise-specific AI responses
Marketplace Ecosystem	Assistants are discoverable, shareable, and reusable across Intel via a central catalog
Enterprise Identity	Intel SSO / Azure AD with AGS entitlements for fine-grained access control

Business Outcomes

- **Democratized AI Access**: Every Intel employee can access enterprise-grade conversational AI
- **Rapid AI Solution Development**: No-code assistant builder enables domain experts to create AI solutions in minutes
- **Knowledge Sharing at Scale**: Marketplace enables reuse of AI assistants across teams and organisations
- **Governed Self-Service**: Private-by-default with compliance gates ensures InfoSecurity policy compliance

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2. Key Capability Patterns

The iGPT platform supports four primary interaction patterns:

Pattern	Description
Conversational	Multi-turn chat with enterprise LLMs. Users interact via natural language to get answers, generate content, analyse data, and complete tasks. Supports model selection and streaming responses.
Assistive	Custom AI assistants with specialised system prompts, configured models, and domain-specific instructions. Built via the no-code Assistant Builder with live testing.
Collaborative	Share assistants across Intel via the Marketplace. Publish as Public (all employees), Secure Groups (Azure AD groups), or keep Private (owner-only).
Knowledge-Augmented	RAG-powered responses grounded in uploaded documents (PDF, DOCX, TXT) and automated data pipelines. Enables context-aware AI within enterprise data.

3. Platform Characteristics

Characteristic	Description
Enterprise Chat AI	Production-ready conversational interface with multiple LLM backends, streaming responses, and conversation history.
No-Code Assistant Creation	Step-by-step wizard for building AI assistants without programming – name, model, temperature, system prompt, test, publish.
Marketplace Ecosystem	Central store for browsing, searching, bookmarking, and running shared assistants across Intel.
Data-Augmented Intelligence	Document upload and pipeline integration enabling RAG-based retrieval for grounded, enterprise-specific responses.
Security by Default	SSO authentication, Azure AD group access, private-by-default assistants, and InfoSecurity compliance enforcement.
Model Flexibility	Multiple LLM options (GPT-4, GPT-3.5, others) with per-session model selection and temperature control via API gateway.

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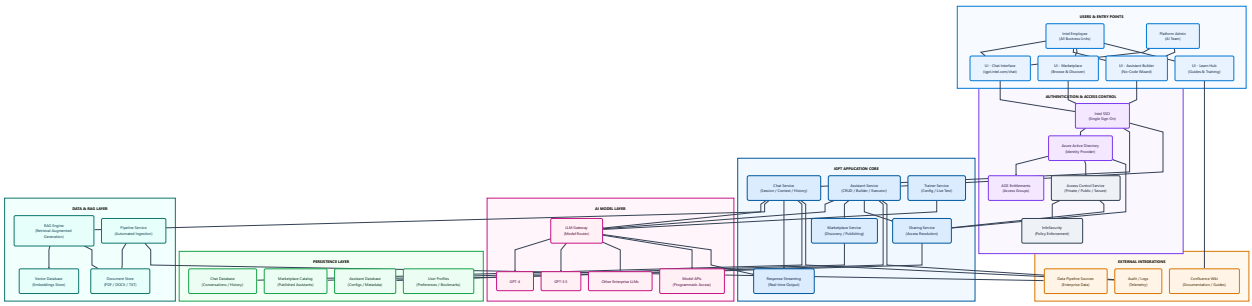
4. Architecture Overview

Architecture Overview – iGPT Platform Component Architecture Diagram



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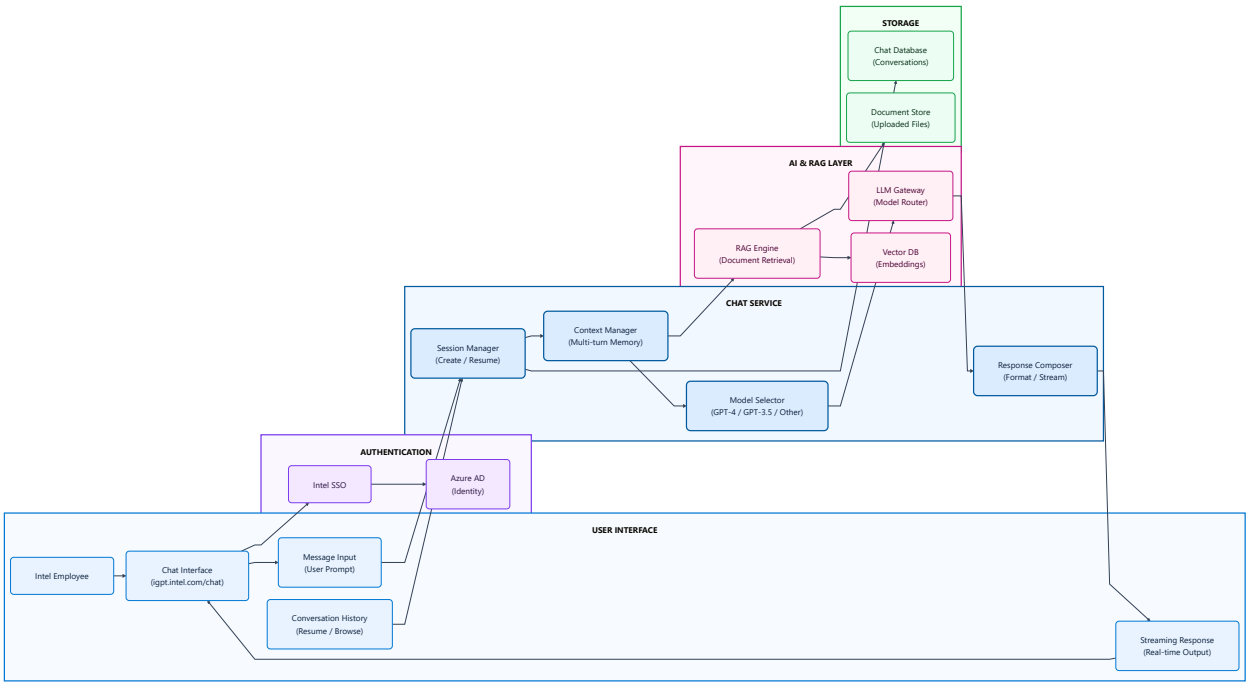
5. Sub-Architecture: Chat & Conversation Flow

Purpose: Detail the end-to-end conversational flow from user input through LLM processing to streaming response delivery.

Chat & Conversation Flow Architecture



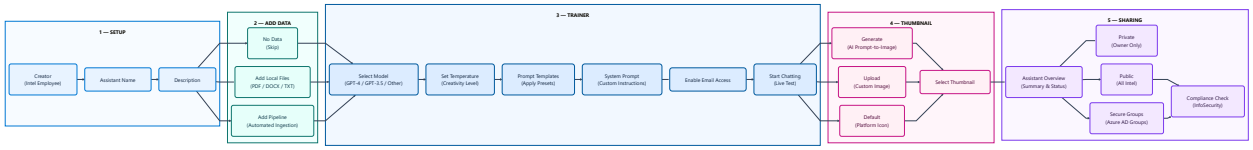
Key Components:

- **Session Manager** – Creates and resumes chat sessions; maintains session state across multiple interactions
- **Context Manager** – Preserves multi-turn conversation memory; passes relevant history to LLM for contextual responses
- **Model Selector** – Per-session model choice (GPT-4, GPT-3.5, others); configurable temperature and parameters
- **RAG Engine** – Retrieves relevant document chunks from Vector DB when assistant has attached data
- **Response Composer** – Formats LLM output and streams tokens in real-time back to the UI

6. Sub-Architecture: Assistant Builder & Lifecycle

Purpose: Detail the 5-tab assistant creation workflow: Setup, Add Data, Trainer, Thumbnail, and Sharing.

Assistant Builder & Lifecycle Architecture



Workflow (5-Tab Builder):

> **Setup** (Name & Description) → **Add Data** (None / Local Files / Pipeline) → **Trainer** (Model Selection, Temperature, Prompt Templates, System Prompt, Email Access, Start Chatting) → **Thumbnail** (Generate / Upload / Default → Select) → **Sharing** (Assistant Overview → Private / Public / Secure Groups)

Key Design Decisions:

- **Private by Default** – All new assistants default to Private (owner-only) access
- **Compliance Gate** – Public and Secure Group publishing requires a compliance checkbox
- **File Restriction** – Assistants with attached files must remain Private (InfoSecurity directive)
- **Live Testing** – "Start Chatting" in the Trainer tab provides real-time testing against the configured LLM
- **AI Thumbnails** – Users can write a prompt to generate a thumbnail image, upload a custom image, or use the platform default

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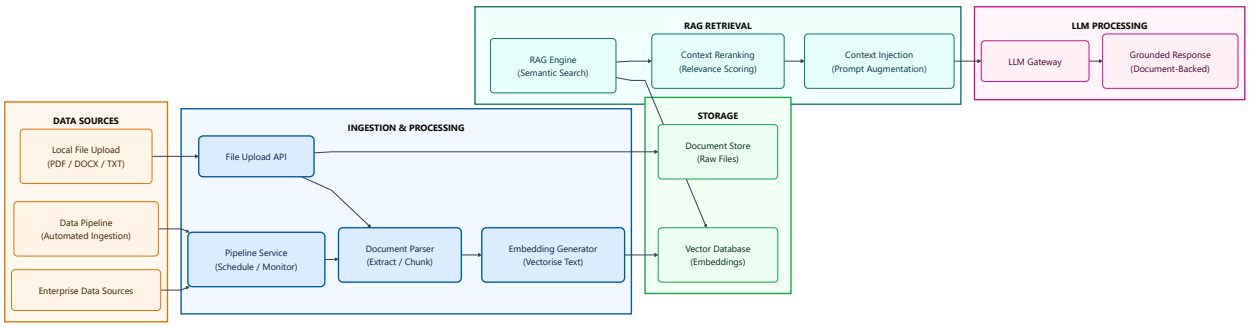
7. Sub-Architecture: Data Management & RAG

Purpose: Detail the data ingestion, document processing, and Retrieval Augmented Generation pipeline.

Data Management & RAG Architecture



Key Components:

- **File Upload API** – Accepts PDF, DOCX, TXT; validates format and size constraints
- **Document Parser** – Extracts text content and chunks documents into retrieval-sized segments
- **Embedding Generator** – Converts text chunks into vector embeddings for semantic search
- **RAG Engine** – Performs semantic similarity search against Vector DB at query time
- **Context Injection** – Augments the user prompt with retrieved document context before LLM processing

Data Restriction: File-attached assistants remain Private per InfoSecurity directive.

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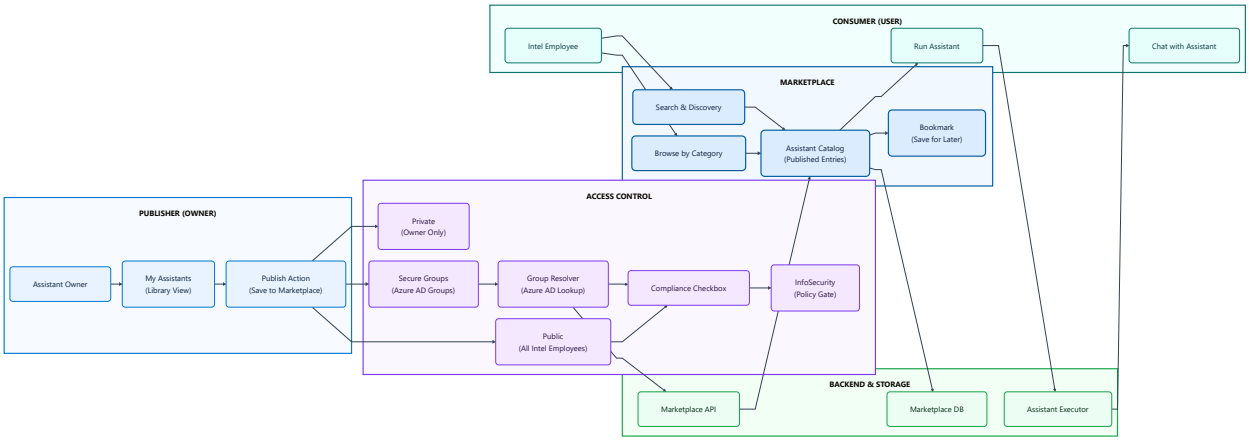
8. Sub-Architecture: Marketplace & Sharing

Purpose: Detail the marketplace ecosystem and access control model for assistant sharing.

Marketplace & Sharing Architecture



Access Control Hierarchy:

> **Private** (Owner Only) → **Secure Groups** (Azure AD Groups via AGS) → **Public** (All Intel Employees)

Key Design Decisions:

- **Owner Visibility** – Owners see sharing status icons in My Assistants but do not see their own assistants in the Marketplace view
- **Azure AD Groups** – Secure Group sharing resolves membership against Azure AD; only Azure AD groups are visible in the iGPT interface
- **Compliance Gate** – Both Public and Secure Group sharing require an explicit compliance checkbox before publishing

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9. Module Capabilities

Chat Interface

Capability	Description
Create Chat	Start new conversation sessions with selected models
Multi-turn Conversations	Maintain context across multiple exchanges
Model Selection	Choose LLM per session (GPT-4, GPT-3.5, others)
Conversation History	Browse, resume, and manage past conversations
Streaming Responses	Real-time token streaming from LLM to UI
Chat with Assistants	Use custom or marketplace assistants in conversation

Assistant Builder

The Assistant Builder follows a 5-tab workflow: **Setup** → **Add Data** → **Trainer** → **Thumbnail** → **Sharing**.

Tab 1: Setup

Capability	Description
Assistant Name	Define a unique name for the assistant
Description	Provide a summary of the assistant's purpose and capabilities

Tab 2: Add Data

Capability	Description
No Data	Skip data attachment; assistant relies solely on LLM knowledge
Add Local Files	Upload PDF, DOCX, TXT files for RAG-based retrieval
Add Pipeline	Configure automated data ingestion from enterprise data sources

Tab 3: Trainer

Capability	Description
Select Model	Choose LLM for the assistant (GPT-4, GPT-3.5, or other enterprise models)
Set Temperature	Adjust creativity level (low = deterministic, high = creative)
Prompt Templates	Apply pre-built prompt templates to guide assistant behaviour
System Prompt	Write custom instructions defining the assistant's persona and rules
Enable Email Access	Grant the assistant access to email context for responses
Start Chatting	Live test the assistant against the configured LLM before saving

Tab 4: Thumbnail

Capability	Description
Generate	Write a prompt to AI-generate a thumbnail image for the assistant
Upload	Upload a custom image as the assistant thumbnail
Default	Use the platform default icon
Select Thumbnail	Preview and select the final thumbnail from generated/uploaded options

Tab 5: Sharing

Capability	Description
Private (default)	Owner-only access; assistant is not visible to others
Public	All Intel employees can discover and run the assistant
Secure Groups	Restrict access to specific Azure AD / AGS entitlement groups
Assistant Overview	Review summary, configuration, and sharing status before publishing
Compliance Check	Required checkbox for Public and Secure Group publishing (InfoSecurity)

My Assistants

Capability	Description
Owned Assistants	View all assistants created by the user
Bookmarked	Access assistants saved from Marketplace
Edit / Delete	Manage owned assistant configurations
Chat	Start conversation with any owned assistant
Status Icons	Visual indicators: Public / Secure / Private / Bookmarked / Owned

Data Management

Capability	Description
No Data	Skip data attachment; rely on base LLM knowledge only
Add Local Files	Upload PDF, DOCX, TXT for RAG-powered retrieval within the assistant
Add Pipeline	Configure automated data ingestion pipelines from enterprise sources
File Formats	PDF, DOCX, TXT, and other document types
File Privacy	Assistants with attached files must remain Private (InfoSecurity directive)

Trainer

Capability	Description
Select Model	Choose the LLM backend for the assistant (GPT-4, GPT-3.5, or other enterprise models)
Set Temperature	Adjust creativity / determinism level for the assistant's LLM responses
Prompt Templates	Apply pre-built prompt presets to accelerate assistant configuration
System Prompt	Define custom instructions, persona, rules, and behavioural guidelines
Enable Email Access	Grant the assistant access to email context for enriched responses
Start Chatting	Live-test the assistant in real-time against the configured LLM before saving

Thumbnail

Capability	Description
Generate	Write a text prompt to AI-generate a visual thumbnail for the assistant
Upload	Upload a custom image file as the assistant thumbnail
Default	Use the platform's default assistant icon
Select Thumbnail	Preview generated / uploaded options and confirm the final selection

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Sharing & Publishing

Sharing Mode	Description
Private (default)	Owner-only access and modification rights
Public	All Intel employees can run; only owner can modify
Secure Groups	Azure AD / AGS groups can run; only owner can modify
Assistant Overview	Summary view of assistant configuration, data, and sharing status before publishing
Compliance Check	Required checkbox acknowledgment before Public or Secure Group publishing

Additional: Compliance checkbox required for Public and Secure Group publishing. File-attached assistants must remain Private.

Marketplace

Capability	Description
Browse	Discover assistants published by other teams
Search & Filter	Find assistants by category and keyword
Bookmark	Save interesting assistants for later access
Run	Execute any accessible marketplace assistant
Publish	Share your assistants with the enterprise

Learn & Resources

Resource	Type
Prompt Formula Training	Video tutorial
Create iGPT Assistants	Step-by-step guide (Confluence)
Share in Marketplace	Publishing guide (Confluence)
Model Licensing	Compliance and licensing information

Model Management

Capability	Description
Model Catalog	Multiple LLM options with descriptions
Temperature Control	Per-session randomness/creativity adjustment
Model APIs	Programmatic access to models
Licensing Info	Model compliance and licensing details

Setup & Configuration

Capability	Description
Setup Wizard	Guided initial configuration for new assistants
Configuration Flow	Multi-step setup workflow
Platform Settings	General platform preferences

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10. External Integrations

Integration	Purpose
Azure Active Directory	Authentication, identity, and group management
Intel SSO	Single sign-on for all Intel employees
AGS Entitlements	Enterprise access group management
Confluence Wiki	Documentation hosting and knowledge base (Learn resources)
LLM APIs	GPT-4, GPT-3.5, and other model API endpoints
Data Pipeline Sources	Automated data ingestion from enterprise systems
InfoSecurity	Policy enforcement for publishing and data compliance

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11. Implementation Phases

Phase 1: Platform Foundation

Purpose: Establish core iGPT infrastructure and enterprise authentication

Deliverable	Description	Status
iGPT Web Application	SPA deployment at igpt.intel.com/chat	✅ Complete
Intel SSO / Azure AD	Enterprise authentication integration	✅ Complete
LLM Gateway	Multi-model routing (GPT-4, GPT-3.5)	✅ Complete
Chat Interface	Multi-turn conversational AI with streaming	✅ Complete
Conversation History	Session persistence and retrieval	✅ Complete

Status Legend: ✅ Complete = Delivered and operational | 🔄 In Progress = Active development | 📋 Planned = Scheduled for future phase

Phase 2: Assistant Builder Rollout

Purpose: Enable no-code custom assistant creation

Deliverable	Description	Status
Assistant Builder (5-Tab Wizard)	No-code creation flow	✅ Complete
Model Selection	Per-assistant LLM selection	✅ Complete
System Prompt Configuration	Custom instruction support	✅ Complete
Temperature & Parameter Controls	Creativity tuning	✅ Complete
Live Testing	Real-time testing in Trainer tab	✅ Complete
Thumbnail Generation	AI-generated and custom thumbnails	✅ Complete

Phase 3: Marketplace Enablement

Purpose: Enable discovery, sharing, and reuse of assistants across Intel

Deliverable	Description	Status
Marketplace Catalog	Browsing, search, and discovery	✅ Complete
Access Control (Private/Public/Secure)	Three-tier sharing model	✅ Complete
AGS Group Integration	Azure AD group-based secure sharing	✅ Complete
Compliance Gate	Required checkbox for public publishing	✅ Complete
Bookmark & Favorites	Save assistants for later	✅ Complete

Phase 4: RAG Pipeline Maturity

Purpose: Enable document-grounded AI responses

Deliverable	Description	Status
File Upload RAG	PDF/DOCX/TXT document grounding	✅ Complete
Document Parser & Chunking	Semantic segmentation pipeline	✅ Complete
Vector Database	Embedding storage and retrieval	✅ Complete
Context Reranking	Relevance scoring for retrieved chunks	🔄 In Progress
Data Pipeline Integration	Automated enterprise data ingestion	🔄 In Progress
RAG Quality Metrics	Precision, accuracy, citation tracking	📋 Planned

Phase 5: Governance Hardening

Purpose: Strengthen governance, monitoring, and operational controls

Deliverable	Description	Status
Audit Logging	Comprehensive telemetry and audit trail	🔄 In Progress
Usage Analytics	Token consumption and user metrics	📋 Planned
Content Safety Controls	Input/output filtering for responsible AI	📋 Planned
Assistant Lifecycle Management	Versioning, deprecation, retirement	📋 Planned
SLA Monitoring & Alerting	Platform health dashboards	📋 Planned

Implementation Timeline

```
Phase 1 [==========] Complete – Platform Foundation
Phase 2 [==========] Complete – Assistant Builder Rollout
Phase 3 [==========] Complete – Marketplace Enablement
Phase 4 [========  ] 80% – RAG Pipeline Maturity
Phase 5 [===       ] 30% – Governance Hardening
```

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12. Platform Stage Lanes

Platform Stage Lanes – Simplified View

This simplified lane view provides an executive-friendly overview of the iGPT platform stages.

Stage	Components	Responsibility	SLA
User Interface	iGPT Web App (SPA), Chat, Builder panels	Frontend Team	99.5% availability
AI Processing	LLM Gateway, Model Routing, Streaming	AI Platform Team	99.5% availability
RAG Pipeline	Document Parser, Embeddings, Vector DB, Context Engine	Data Engineering	99.0% availability
Marketplace	Catalog, Discovery, Publishing, Access Control	Platform Team	99.5% availability
Identity & Access	Intel SSO, Azure AD, AGS Entitlements	Identity Team	99.9% availability
Storage	Document Store, Vector DB, Marketplace DB, Conversation History	Infrastructure Team	99.9% availability

Stage Summary

Stage	Components	Key Metric
Sources	User input, uploaded documents, data pipelines	Ingestion success rate
Processing	LLM Gateway, RAG Engine, context injection	Response latency P95
Storage	Vector DB, Document Store, Marketplace DB	Storage availability
Serving	Chat responses, assistant execution, marketplace catalog	User satisfaction
Governance	Audit logs, compliance gates, access control	Audit completeness

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13. TOGAF BDAT Crosswalk

TOGAF BDAT Crosswalk – Conceptual Mapping

This conceptual mapping aligns the iGPT architecture to TOGAF Business, Data, Application, and Technology (BDAT) domains, supporting enterprise architecture governance.

Business Architecture (B)

Capability	Description	Value Delivered
Conversational AI	Natural language chat with enterprise LLMs	Instant AI access for all employees
Custom Assistant Creation	No-code AI solution development	Rapid domain-specific AI solutions
Marketplace Ecosystem	Assistant sharing and discovery	Knowledge reuse at enterprise scale
Document-Grounded Q&A	RAG-powered contextual responses	Accurate, enterprise-specific answers

Data Architecture (D)

Layer	Purpose	Quality Standard
Document Store	Raw uploaded files (PDF/DOCX/TXT)	Validated format and size
Vector Database	Embeddings for semantic search	Indexed, refreshed on upload
Marketplace Database	Assistant catalog and metadata	Governed, searchable
Conversation History	Session persistence	User-scoped, encrypted

Application Architecture (A)

Application	Role	Integration Pattern
iGPT Web Application	User interface (SPA)	REST API to backend
LLM Gateway	Model routing and management	API calls to LLM providers
RAG Engine	Document retrieval and grounding	Vector similarity search
Marketplace Service	Catalog and access control	REST API with AGS
Assistant Builder	No-code creation wizard	Multi-step wizard UI

Technology Architecture (T)

Technology	Purpose	Provider
Azure AD	Identity management	Microsoft
AGS	Enterprise entitlements	Intel
LLM APIs	GPT-4, GPT-3.5, other models	OpenAI / Azure OpenAI
Vector Database	Embedding storage	Platform-managed
Azure Infrastructure	Compute and storage	Microsoft Azure

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14. Ownership & Operating Model

Ownership / Operating Model

This view maps iGPT platform components to responsible teams, supporting RACI discussions and operational governance.

RACI Matrix

Component	Frontend Team	AI Platform	Data Engineering	Identity Team	Platform Team	Governance
iGPT Web Application	**R/A**	I	I	I	C	I
LLM Gateway	C	**R/A**	I	I	C	I
Chat Interface	**R/A**	C	I	I	I	I
Assistant Builder	**R/A**	C	I	I	C	I
RAG Engine	I	C	**R/A**	I	I	I
Vector Database	I	I	**R/A**	I	C	I
Marketplace Service	I	I	I	I	**R/A**	C
Intel SSO / Azure AD	I	I	I	**R/A**	I	I
AGS Entitlements	I	I	I	**R/A**	C	I
Audit Logging	I	C	I	I	C	**R/A**
Compliance Gate	I	I	I	I	C	**R/A**
InfoSecurity Policies	I	I	I	I	I	**R/A**

RACI Legend:

Code	Role	Definition
R/A	Responsible & Accountable	Owns delivery and final decision authority
C	Consulted	Provides input before decisions are made
I	Informed	Notified after decisions are made

Support Model

Tier	Scope	Team	SLA
L1	User issues, access requests, basic chat issues	Service Desk	4 hours
L2	Assistant builder issues, marketplace problems	Platform Team	8 hours
L3	LLM failures, RAG pipeline issues, performance	AI Platform / Data Engineering	24 hours
L4	Architecture, design decisions, governance	Enterprise Architecture	5 days

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15. L2 Architecture Deliverables

Identity & Access Architecture

- Configure **Intel SSO / Azure AD** as primary identity provider for all iGPT users
- Implement **AGS Entitlements** for fine-grained group-based access control on Secure Group sharing
- Define role mappings for platform administration, assistant creation, and marketplace publishing
- Establish access audit logging for all authentication and authorization events

Integration Architecture

- Define **LLM Gateway** routing configuration for multi-model support (GPT-4, GPT-3.5, others)
- Configure **Data Pipeline** integration for automated enterprise data ingestion
- Establish **Marketplace API** specifications for assistant publishing, discovery, and execution
- Define **RAG Engine** integration patterns for vector storage and retrieval

Security Architecture

- Implement **Private-by-default** access control for all new assistants
- Configure **Compliance Gate** workflow for Public and Secure Group publishing
- Enforce **File Privacy** directive (assistants with files remain Private)
- Define data classification and handling policies for uploaded documents
- Establish content safety controls for LLM input/output filtering
- Configure comprehensive audit logging and telemetry collection

Platform Architecture

- Define **Assistant Builder** lifecycle management (create, configure, test, publish, retire)
- Establish **Marketplace** catalog governance (quality standards, metadata requirements)

- Configure **LLM Gateway** performance monitoring (latency, error rates, token consumption)
- Define **RAG Pipeline** quality metrics (retrieval precision, answer accuracy, citation accuracy)
- Establish platform capacity planning and scaling policies

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16. Governance & Operations

Governance Framework

AI Governance

Domain	Owner	Controls
Model Approval	AI Platform Team	LLM model registry, licensing compliance
Content Safety	AI Platform Team	Input/output filtering, responsible AI policies
Publishing Approval	Governance	Compliance gate for Public/Secure Group sharing
Audit & Evidence	Compliance	Comprehensive logging and telemetry
Data Privacy	InfoSecurity	File privacy directive, data classification

Access Governance

Feature	Description	Enforcement
Intel SSO	Enterprise single sign-on authentication	All users, all sessions
Azure AD	Identity provider; group management	User/group resolution
AGS Entitlements	Fine-grained group-based access	Secure Group sharing
Private by Default	All new assistants default to owner-only	Assistant creation
File Privacy	Assistants with files remain Private	InfoSecurity directive
Compliance Gate	Required checkbox before broader sharing	Publishing workflow
Audit / Logs	Telemetry and audit trail	All platform operations

> **InfoSecurity Directive:** "Assistants with Files can only be published as Private. This is not a technical restriction, but a direction from InfoSecurity."

Operational Procedures

Monitoring & Alerting

Metric	Threshold	Action
LLM Gateway error rate	> 2%	Page on-call AI Platform
Chat response latency P95	> 5 sec	Investigate model performance
RAG retrieval precision	< 80%	Review RAG pipeline tuning
Marketplace publishing failures	> 1%	Alert Platform Team
Token consumption	> 80% monthly budget	Capacity planning alert

Incident Management

Severity	Definition	Response Time	Resolution Time
P1	Platform outage (chat/marketplace down)	15 min	4 hours
P2	LLM degradation or RAG failures	1 hour	8 hours
P3	Assistant builder issues, minor bugs	4 hours	24 hours
P4	Enhancement request	1 week	Sprint planning

Change Management

Change Type	Approval	Lead Time
Model addition/update	AI Platform + Governance	10 days
Platform feature release	Platform Team	5 days
Security policy update	Governance + InfoSecurity	5 days
Emergency (security)	Emergency CAB	2 hours

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17. Architectural Flows

Chat Conversation Flow

```
User → iGPT Web App → Intel SSO (Authentication) →
Chat Interface → Message Input → LLM Gateway (Model Routing) →
Selected LLM (GPT-4/GPT-3.5) → Streaming Response →
Response Rendering → Conversation History (Persist) → User
```

RAG-Grounded Chat Flow

```
User → Chat Interface → Message Input →
RAG Engine (Semantic Search) → Vector DB (Embedding Retrieval) →
Context Reranking (Relevance Scoring) →
Context Injection (Prompt Augmentation) →
LLM Gateway → Selected LLM → Grounded Response with Citations → User
```

Assistant Creation Flow

```
Creator → Assistant Builder:
  Tab 1 (Setup): Name + Description →
  Tab 2 (Add Data): None / Local Files / Pipeline →
  Tab 3 (Trainer): Model Selection → Temperature → System Prompt → Live Test →
  Tab 4 (Thumbnail): Generate / Upload / Default →
  Tab 5 (Sharing): Private (default) / Public / Secure Groups →
  Compliance Gate (if Public/Secure) → InfoSecurity Check →
  Save to Marketplace DB → Published
```

Marketplace Publishing Flow

```
Assistant Owner → My Assistants (Library) → Publish Action →
Access Control Selection:
  Private → Save (no gate)
  Public → Compliance Checkbox → InfoSecurity Gate → Marketplace Catalog
  Secure Groups → AGS Group Resolver (Azure AD Lookup) → Compliance Checkbox → InfoSecurity Gate → Marketplace Catalog

Consumer → Browse / Search → Marketplace Catalog →
Run Assistant → Assistant Executor → Chat with Assistant
```

Document Ingestion Flow

```
User → File Upload API (PDF / DOCX / TXT) → Format Validation →
Document Parser (Extract / Chunk) → Embedding Generator (Vectorise) →
Vector Database (Store Embeddings) + Document Store (Store Raw Files)

Data Pipeline → Pipeline Service (Schedule / Monitor) →
Document Parser → Embedding Generator → Vector Database
```

Authentication & Access Flow

```
User → iGPT Web App → Intel SSO (Azure AD) →
Token Validation → Session Establishment →
AGS Entitlements Check (for Secure Group content) →
Access Granted / Denied → Platform Features
```

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18. Related Architecture Patterns

iGPT is one of **three complementary AI architecture patterns** at Intel. Each pattern is purpose-built for a distinct class of use cases and maintains its own runtime, identity model, and governance boundaries.

Pattern	Scope	Entry Point
SAP Joule	SAP transactions, navigation, analytics, SAP knowledge	Joule Assistant bar in SAP apps
ECA / Azure Copilot Studio	Enterprise data analytics, agentic workflows, non-SAP integrations	Copilot Agent UX (Teams / MS365)
iGPT Platform (this document)	General-purpose chat, custom assistants, marketplace, document Q&A	igpt.intel.com/chat

High-Level Integration Points

- **iGPT → Joule / ECA**: No direct integration today. iGPT operates as a standalone self-service platform. Users context-switch to Joule for SAP needs or Copilot Studio for ECA data queries.
- **Future Roadmap**: MCP/API integration may enable iGPT assistants to be registered as tools within ECA/Azure agentic workflows, providing a bridge between self-service AI and governed enterprise data.

When to Use iGPT vs Other Patterns

If you need…	Use iGPT	Use Joule or ECA Instead
General-purpose AI chat	✅	—
Custom assistant for your team (no-code)	✅	—
Document Q&A from uploaded files	✅	—
Sharing assistants via Marketplace	✅	—
SAP transaction processing	—	✅ SAP Joule
Enterprise data analytics (Snowflake/Databricks)	—	✅ ECA / Azure
Multi-system agentic workflows	—	✅ ECA / Azure

Selection Guidance

For detailed use case routing, decision frameworks, and guidance on choosing between the three patterns, refer to:

> 📄 **AI Architecture Selection Guide** — `AI-Architecture-Selection-Guide.md`

Related Documents

Document	Location
ECA & AI Enablement Architecture	`ECA_AI_Enablement_Architecture_DocumentV1.md`
SAP Joule Architecture Overview	`Current AI Architecture Options/SAP-Joule-Architecture-Overview.md`
AI Architecture Selection Guide	`AI-Architecture-Selection-Guide.md`

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Appendix A: Component Glossary

Component / Abbreviation	Description
iGPT	Intel Generative Pre-trained Transformer – the enterprise AI platform
LLM	Large Language Model – AI models trained on large text corpora (e.g. GPT-4, GPT-3.5)
RAG	Retrieval Augmented Generation – technique that augments LLM prompts with retrieved document context for grounded responses
SPA	Single Page Application – web application architecture that loads a single HTML page and dynamically updates content without full page reloads
SSO	Single Sign-On – one-time authentication granting access to multiple enterprise systems
API	Application Programming Interface – a set of protocols enabling software components to communicate
CRUD	Create, Read, Update, Delete – the four basic operations for data management
UI	User Interface – the visual front-end through which users interact with the platform
DB	Database – structured data storage system
AD	Active Directory – Microsoft's directory service for identity and access management (see Azure AD)
Azure AD	Azure Active Directory – Microsoft cloud-based identity and access management service
AGS	Access Group Service – Intel's enterprise entitlements service for fine-grained group-based permissions
LLM Gateway	Model routing layer that directs requests to the appropriate LLM backend
Vector Database	Stores document embeddings as numerical vectors for semantic similarity search
Assistant Builder	No-code wizard for creating custom AI assistants
Marketplace	Enterprise store for browsing, sharing, and running AI assistants
InfoSecurity	Intel's information security team and policy enforcement function
Confluence	Atlassian wiki platform used for Intel documentation and knowledge bases
GPT-4 / GPT-3.5	OpenAI large language models available via the LLM Gateway
PDF	Portable Document Format – Adobe file format for fixed-layout documents
DOCX	Microsoft Word document format (Office Open XML)
TXT	Plain text file format

Component / Abbreviation	Description
TB / LR	Top-to-Bottom / Left-to-Right – Mermaid flowchart direction indicators used in architecture diagrams

Generated on March 2, 2026

Source: iGPT platform analysis (https://igpt.intel.com/chat)

Modules: 10 | Features: 55 | Security Features: 6 | Integrations: 7

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Document Control

Version	Date	Author	Changes
1.0	March 2026	Sajiv Francis	Initial release
1.1	March 2026	Sajiv Francis	Added Related Architecture Patterns section (three-pattern model: Joule, ECA/Azure, iGPT); cross-references to Selection Guide

Classification: Internal Use

Review Cycle: Quarterly

Next Review: June 2026

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iGPT Architecture Overview

End of Document